

Laboratory of Translational Oncology
Medical University of Gdańsk
Dębinki 1
80-211 Gdańsk, Poland

April 24, 2026

The Editorial Office
BMC Bioinformatics
BMC, part of Springer Nature

Dear Editors,

We are pleased to submit our manuscript entitled “*Classical Machine Learning Outperforms and Excels in Calibration Compared to Single-Cell Large Language Models for Cell-Type Annotation at Scale*” for consideration as a Research Article in **BMC Bioinformatics**.

Single-cell large language models (scLLMs) such as scGPT, Geneformer, and scFoundation have dominated recent methodological discourse in single-cell genomics, on the premise that pre-training on tens of millions of cells yields transferable representations that surpass classical task-specific classifiers. However, the field’s evaluation of these models has focused almost exclusively on point-estimate accuracy, while two arguably more important dimensions—*statistical calibration* (whether confidence scores are trustworthy) and *matched-control performance* (whether scLLMs beat classical baselines under truly fair comparisons)—have been largely ignored.

Our contribution addresses these gaps directly. Across six diverse public single-cell atlases totalling 1.3 million cells (triple-negative breast cancer, Indonesian PBMC, multi-region brain, multi-tissue tumour microenvironment, and human and pig pancreas), we benchmark four families of cell-type annotation pipelines under uniform conditions:

1. the standard scGPT frozen-embedding pipeline with a Logistic Regression head;
2. the same scGPT embeddings paired with stronger non-linear probes (XGBoost, Random Forest);
3. three classical learners (LogReg, XGBoost, RF) on highly variable genes restricted to scGPT’s 60,697-token vocabulary—the matched-control we believe is missing from prior scLLM benchmarks; and
4. full end-to-end fine-tuning of scGPT on triple-negative breast cancer.

All comparisons report Macro F1 with non-parametric bootstrap 95% confidence intervals (1,000 resamples) and Expected Calibration Error (ECE) with reliability diagrams.

Our principal findings are: (i) classical Logistic Regression on vocabulary-matched HVGs reaches Macro F1 = 0.94–0.99 on all six atlases with $\text{ECE} \leq 0.013$, outperforming or matching every scGPT-based pipeline; (ii) the standard scGPT + LogReg pipeline is systematically miscalibrated.

brated (ECE 0.07–0.18) and substantially less accurate (Macro F1 0.27–0.71)—a phenomenon we characterise as *single-cell hallucination*; and (iii) end-to-end fine-tuning of scGPT on TNBC reaches Macro F1 = 0.975 / ECE = 0.003, statistically indistinguishable from the much simpler scGPT + XGBoost pipeline (0.972) and still below the vocabulary-matched LogReg baseline (0.993). Importantly, we also demonstrate that scGPT’s attention-derived gene representations recover known co-regulatory structure in a fully zero-shot setting, indicating that the model retains value for label-free exploratory analyses—a balanced view we believe will be useful to the community.

We believe this work fits the scope of *BMC Bioinformatics* for several reasons. The journal has a long-standing focus on rigorous benchmarking of computational methods in genomics; our study is, to our knowledge, the first systematic, calibration-aware benchmark of an scLLM at this scale and under these matched controls. The conclusions are immediately actionable for any laboratory choosing an annotation pipeline for a new atlas, and they argue concretely for the inclusion of calibration metrics and vocabulary-matched baselines as standard reporting requirements in future scLLM publications.

Reproducibility has been a first-class concern throughout. The full analysis code, end-to-end fine-tuning script, raw per-model prediction files, bootstrap CSV outputs, and reliability-diagram source data are openly available at <https://github.com/alirezakhosravii/scGPTvsClassical> under an MIT licence; a permanent DOI-pinned snapshot will be archived on Zenodo upon acceptance. All datasets are public, with direct accession URLs listed in the manuscript.

This work has not been published previously, is not under consideration elsewhere, and all listed authors have approved the submission. The authors declare no competing interests.

We thank you for considering our manuscript and look forward to your response.

Sincerely yours,

Anna Supernat, PhD
Corresponding Author
Laboratory of Translational Oncology
Medical University of Gdańsk
abednarz@gumed.edu.pl